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TERM END EXAMINATIONS (TEE) –JULY 2026

Programme	:	**B.Tech**	**Semester**	:	**Winter Semester 2025-26**
Course Name	:	**Electric Circuits and Systems**	**Course Code**	:	**EEE1001**
Date/Session:	:	**1st July 2026**	**Slot**	:	**B11+B12+B13**
Time	:	**3 Hrs.**	**Max. Marks**	:	**100**

Answer ALL the Questions

Q. No.		Question Description	Marks
		PART A – (60 Marks)	
1	(a)	What resistance should be connected across x-y in the circuit shown in Fig. 1 such that maximum power is developed across this load resistance? What is the amount of this maximum power? Also, calculate the maximum power transfer efficiency.	12

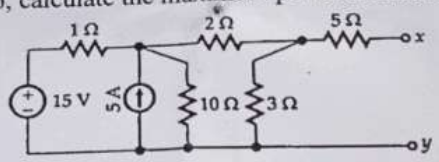


Fig. 1

OR

	(b)	Apply Norton's theorem to determine the current flowing through the resistance of 6 Ω connected across the terminals A and B in Fig. 2. Also calculate the potential of point A. What will be the current through the 6 Ω resistor across AK?	12

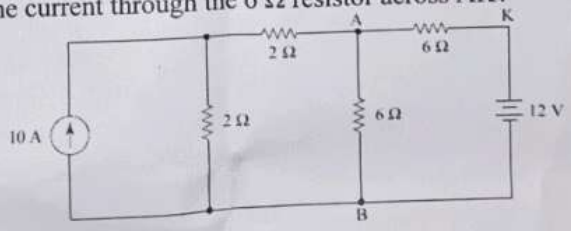


Fig. 2

2	(a)	Calculate the total current, power, and power factor of the whole circuit in Fig. 3. Also calculate the reactive power and apparent power of the circuit.	12

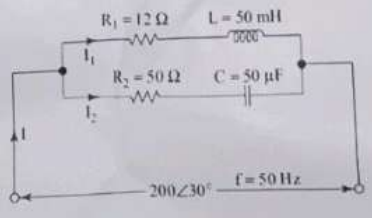


Fig. 3

		OR	
	(b)	The switch in the circuit in Fig. 4 has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.	12

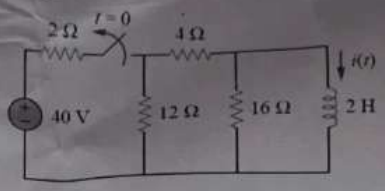


Fig. 4

3	(a)	(i) Explain the working principle of a DC motor with a neat sketch. (ii) Explain the working principle of an induction motor with a neat sketch.	2×6
		OR	
	(b)	(i) Explain the working principle of a transformer with a neat sketch. (ii) Explain, with neat sketches, the two different types of core constructions in a typical transformer.	2×6
4	(a)	(i) Explain in brief the formation of a depletion region in a p-n junction. (ii) What are Zener breakdown and Avalanche breakdown in a Zener diode? (iii) Draw the VI characteristic of a typical p-n junction diode. Also, label and explain the significant regions in it.	3×4
		OR	
	(b)	Name the three different configurations of bipolar junction transistors. Present neat sketches of the circuits for each type of configuration (for both n-p-n and p-n-p types).	12
5	(a)	Explain full-adder with its truth table, Boolean expressions, and logic circuit diagram.	12
		OR	
	(b)	Explain S-R flip-flop with its block diagram, truth table, characteristic table, excitation table, characteristic equation, and logic circuit diagram.	12
		PART B – (40 Marks)	
6		Two batteries of EMF 2.05 V and 2.15 V having internal resistances of 0.05 Ω and 0.04 Ω, respectively are connected together in parallel to supply a load resistance of 1 Ω as shown in Fig. 5. Calculate using the superposition theorem, current supplied by each battery and also the load current.	8

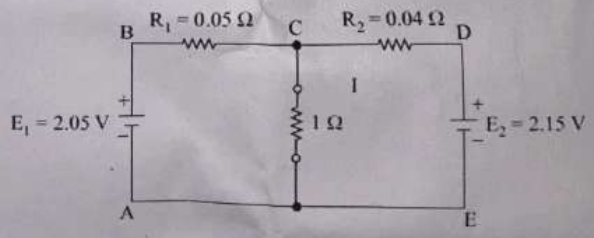
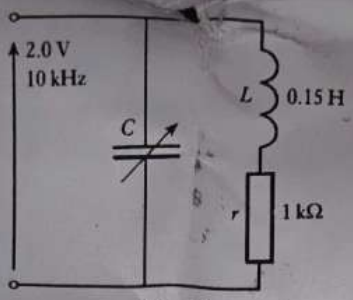


Fig. 5

7		A variable capacitor has been connected in parallel to a coil with an inductance of 0.15 H and resistance of 1 kΩ across a 2 V, 10 kHz AC supply as shown in Fig. 6. Calculate:	8

		(a) the capacitor capacitance when the supply current is minimum; (b) the impedance of the circuit under resonance; and (c) the supply current under resonance.  Fig. 6	
8		Two coils of number of turns $N_1 = 1000$ and $N_2 = 400$, respectively, are placed near each other. They are magnetically coupled in such a way that 75% of the flux produced by the one of 1000 turns links the other. A current of 6 A produces a flux of 0.8 mWb in N_1 and the same amount of current produces a flux of 0.5 mWb in the coil of N_2 turns. Determine L_1, L_2, M, and K for the coils, where the symbols denote their usual meaning.	8
9		A half-wave rectifier circuit has been made using a step-down transformer of turn ratio 10:1. The input voltage is $v = 325\sin\omega t$ V and the diode forward resistance is 25 Ω. A load resistance of 1.2 kΩ has been connected in the circuit. Assuming a secondary winding resistance of the transformer as 1Ω, calculate the following: (a) rms value of load current (b) rectification efficiency, and (c) ripple factor.	8
10		Convert the following bases as instructed (show the conversions in detailed steps): (i) $(0.8125)_{10} = (?)_2$ (ii) $(0.111\ 001110)_2 = (?)_8$ (iii) $(732.14)_8 = (?)_{10}$ (iv) $(1111\ 1110\ 0011)_2 = (?)_{16}$	4×2

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